

Participant Handbook

Foundations of Clinical Research · Level 1: From Clinical Question to Publication

Provider: Neudata · #ClearDataClearImpact

Welcome

Every day you make clinical decisions that research could improve. Over eight weeks, with one session every Saturday, this course takes you through the whole research process: from a question at the bedside, to a sound study design, to analysing data, to writing and publishing the paper.

You do not need any research or statistics background. We go slowly, with pictures, examples from your own wards and a lot of group work. You will never be asked to calculate anything beyond a simple 2×2 table, and you do not need any software: the trainers run the analyses, and you learn to read and write them up.

The course at a glance

Week	Session (all on Saturday)	Theme	Your group builds
1	1	ASK: from clinical problem to research question	Proposal: topic, PICO, question, aims
2	2	FIND & READ: searching, managing references, reading a paper in 15 minutes	Proposal: search, 5 references, problem statement, working title
3	3	DESIGN: choosing the right study	Proposal: design, setting, population
4	4	BIAS & EFFECT: bias, confounding, RR/OR/NNT, diagnostic tests, forest plots	Proposal: exposure, outcomes, bias plan
5	5	DATA: CRFs, data management, describing data, <i>meeting the course dataset</i>	Proposal: variables, CRF, data plan · Manuscript: Table 1 understood
6	6	ANALYSE: CIs, p-values, tests, regression, survival, missing data, <i>analysing the course dataset</i>	Proposal: sample size, analysis plan · Manuscript: results interpreted
7	7	PROTECT & WRITE: ethics, registration, integrity, and writing Methods and Results	Proposal: ethics, feasibility, publication plan · Manuscript: Methods + Results

Week	Session (all on Saturday)	Theme	Your group builds
8	8	PUBLISH: Introduction, Discussion, abstract, journals, submission, peer review and presentations	Manuscript: complete draft · Proposal presented

Each session lasts 3 hours with a 15-minute break. It has a short recap, two teaching blocks with frequent "check your understanding" moments, and time to build your proposal or paper. Each session also has extra slides at the end for anyone who wants to read further at home.

The running example

One clinical topic travels with us: **fluids in septic shock**. In Weeks 1–4 we ask, search and design the question, and look at two real papers (a randomised trial and a cohort study). In Weeks 5–8 we analyse FLUID-ICU, a realistic *simulated* dataset of 620 ICU patients, and write it up as a paper.

Your group and its two products

You stay in the same group of about six for all eight sessions. Agree roles in Session 1 (they can rotate): facilitator, scribe, timekeeper, literature lead, methods & data lead, presenter.

1. **A research proposal on your own question**, built one part every two weeks and presented in Session 8 (6 minutes + 2 minutes of feedback). See [Assignment/proposal_brief.md](#).
2. **A draft manuscript from the course dataset**. Each group answers a different question on FLUID-ICU (e.g. fluid balance and mortality, kidney injury, ventilator-free days, lactate and missing data, ICU length of stay), using its results pack. You draft Methods and Results in Session 7 and the rest in Session 8. After the course, groups review each other's drafts and reply like real authors. See [Assignment/manuscript_brief.md](#) and [Assignment/peer_review_instructions.md](#).

Between sessions (one week; about 30–60 minutes)

After	Do this
Session 1	Think of synonyms for your PICO terms
Session 2	Finish Proposal Part 1 and the literature matrix; read the abstract of Paper 2
Session 4	Finish Proposal Part 2
Session 5	Read your group's results pack (Table 1 and figures)
Session 6	Finish Proposal Part 3; read Papers 1 and 2; start your 6-slide presentation
Session 7	Finish Proposal Part 4 and your presentation; re-read your Methods + Results
Session 8	Polish the manuscript; peer-review swap and response letter within 2 weeks

What you need

A pen, a notebook, and (optional) a laptop or phone for PubMed and the online quiz. No software to install. Participants who have done the free Neudata course *Clinical Data Analysis in R* may re-run the course analyses with the scripts in `Data/R/`.

Your toolkit

Resource	File
Research proposal template	Templates/Research_Proposal_Template.docx
Study protocol template	Templates/Study_Protocol_Template.docx
Informed consent template	Templates/Informed_Consent_Template.docx
CRF + data dictionary template	Templates/CRF_Template.docx
Literature review worksheet	Templates/Literature_Review_Worksheet.docx
Statistical analysis plan template (+ worked example <code>Data/SAP_FLUID-ICU.md</code>)	Templates/Statistical_Analysis_Plan_Template.docx
Manuscript (IMRaD) template	Templates/Manuscript_Template_IMRaD.docx
Cover letter and response-to-reviewers templates	Templates/Cover_Letter_Template.docx, Templates/Response_to_Reviewers_Template.docx
Conference abstract & poster guide	Templates/Conference_Abstract_Template.docx
Critical appraisal checklist (10 questions)	Resources/Critical_Appraisal_Checklist.pdf
Risk-of-bias tool chooser	Resources/Risk_of_Bias_Tool_Chooser.pdf
Writing & submission checklist	Resources/Writing_and_Submission_Checklist.pdf
Sample ethics application, sample grant concept note (fictional)	Resources/
2×2 and sample-size calculators	Resources/tools/
Course dataset, data dictionary, results packs	Data/
English–Vietnamese glossary, further reading	References/

The quiz

In Session 1 and again at the end of Session 8 you answer the same 24 questions. It is anonymous (participant number only) and measures what the *course* achieved, not you.

After the course

Take your proposal to the ethics committee and keep improving your manuscript. This course is Level 1. If you want to run analyses yourself, Level 2 is a hands-on course in R (prerequisite: this course and the free *Clinical Data Analysis in R* course). Level 3 offers advanced, specialised methods on request (`Course_Levels/`).

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